

PSP · HLE · RECOMPILATION

roadmap.**dissidia_recomp()**

// Phased implementation plan: threads → file I/O → graphics → audio → input



```
$ psprecomp --game "Dissidia 012" --phases 5
[1/5] Thread/Kernel bootstrap ... nearly done
[2/5] File I/O ... unlocks assets
[3/5] Graphics (GE processor) ... the big one
[4/5] Audio ... stubbable
[5/5] Input ... straightforward
```

Phase 1 – Thread/Kernel bootstrap // you're here, nearly done

- ▶ `sceKernelCreateThread`, `StartThread`, `ExitThread`, `GetThreadId`
- ▶ `sceKernelAllocPartitionMemory` / `GetBlockHeadAddr` (heap)
- ▶ `sceKernelSetCompiledSdkVersion` variants (just return 0)
- ▶ `sceKernelGetSystemTimeLow` (return host clock)
- ▶ `sceKernelCreateCallback`, semaphores, event flags

Phase 2 – File I/O

- ▶ `sceIoOpen`, `sceIoRead`, `sceIoClose`, `sceIoLseek`
- ▶ Map PSP file paths (`disc0:/PSP_GAME/USRDIR/`) to your local folder
- ▶ This unlocks asset loading and gets you past most early crashes

Phase 3 – Graphics // the big one

- ▶ PSP game code calls `sceGuXxx` to build GE display lists in RAM
- ▶ You don't emulate `sceGu` calls – they're recompiled PSP code
- ▶ You emulate the **GE command processor**: parse the display list, translate to Vulkan/DX12/OpenGL draw calls
- ▶ References: [AljandrOrtega's SDL3+Vulkan renderer](#) (from-scratch, MIT), [psp-web-recomp](#) (WebGL2), both based on PPSSPP's GE semantics as oracle

Phase 4 – Audio

- ▶ `sceAudio` → raw PCM output (SDL2/miniaudio)
- ▶ `sceSasCore` → 32-channel ADPCM/ADSR mixer; [uofw](#) has a clean reference implementation
- ▶ ATRAC3+ music → can be stubbed to silence initially, full decode needs `libatrac3plus`

Phase 5 – Input

- ▶ `sceCtrlReadBufferPositive` → map SDL2/XInput gamepad buttons to PSP button bits

Most reusable existing work

- ▶ [OceanCrab](#) – SpongeBob PSP, uses PSPRecomp (same toolchain as yours), most directly relevant
- ▶ [psp-web-recomp](#) – most complete HLE layer (threads, memory, file I/O, save data, controller, audio, movie skipping)
- ▶ [nakagawa-recomp](#) – PAC-MAN CE reached menu, shows NID aliasing pattern

The shortcut for Dissidia specifically

*// tip: once Phase 2 (file I/O) is working, the game will likely run all its init code and reach the GE display list submission. At that point the output is just a memory buffer – you can dump it and compare against PPSSPP frame-by-frame to validate your GE interpreter *before you write a single Vulkan line.**

Sources

- ↳ [OceanCrab](#) (SpongeBob PSP, PSPRecomp)
- ↳ [psp-web-recomp](#) (WebAssembly, most complete HLE)
- ↳ [AljandrOrtega](#) PSP recompilation (SDL3+Vulkan renderer)
- ↳ [nakagawa-recomp](#) / PAC-MAN CE PR
- ↳ [PSPRecomp architecture doc](#)